AIMScribe v3 — Server Deployment

Document 2 of 2 — for CMED's server and infrastructure team · The machine, the gateway, the storage and the monitoring · AIMS LAB, United International University · 16 October 2026

	The two documents
1. The other	**Software integration** — the four APIs, the key, the fields, the data model. **For the team writing the code**
2. This one	**Server deployment** — where the server runs, how it is exposed, how to tell it is healthy, where the audio lives, and what changes on CMED's side. **For the team working on infrastructure**

What is being deployed, in one paragraph
The whole AIMS LAB server — nine services, two databases, the HTTPS gateway and the archive worker — moves onto **one DigitalOcean droplet in Bangalore**. The recordings themselves are not kept on it: they are uploaded to **Cloudflare R2** and removed. **Nothing CMED runs needs to change except one configuration line**, and the clinic PCs need no firewall rule at all.

The three things this document exists to tell you
1. Which endpoints work **today** and which only work **after** deployment — section 1, so nobody waits on the wrong thing.
2. CMED changes exactly one line when we go live: `AIMS_SERVER_URL` in your own server environment. Your web page changes nothing, because it never knew our address — section 10.
3. The clinic PCs need no inbound port, no port forward and no firewall change, on CMED's side or the clinic's — section 9.

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PART A

Endpoint status

What works today, what works after deployment

1 · Every endpoint, before and after

"Today" means the system as it runs now, on a development machine on a private network. "After" means once the droplet is live.

	Endpoint	Today	After	Why
1	`GET https://<host>/health`	**NO**	**YES**	Needs a public name and a certificate. Both arrive with the droplet
2	`GET https://<host>/`	**NO**	**YES**	same
3	`POST …/api/v2/clinical/patient-information` *API 2*	**NO**	**YES**	**The code is finished and tested.** It has had no address to answer on
4	`POST …/api/v2/clinical/prescription` *API 3, Channel B*	**NO**	**YES**	same as 3
5	`ws://127.0.0.1:5050/ws` → `start` *API 1*	**YES**	**YES**	**Unaffected.** Loopback on the clinic PC — our server was never involved
6	`ws://…` → `prescription_built`	**YES**	**YES**	unaffected
7	`ws://…` → `doctors`	**YES**	**YES**	unaffected
8	`ws://…` → `status`	**YES**	**YES**	unaffected
9	`GET …/api/v2/clinical/patient/{id}/visits` *ask what we hold — new*	**NO**	**YES**	same as 3. Without the key: **401**
10	`POST …/api/v2/session/reassign` *recorders only — new*	—	**YES**	**Not CMED's.** Kept for older recorders. Without a device credential: **401**
12	`POST …/api/v2/session/marker` *recorders only — new*	—	**YES**	**Not CMED's.** Which patient was opened, or had a prescription built, at which second of a recording. Without a device credential: **401**
11	`POST …/api/v2/device/health` *recorders only — new*	—	**YES**	**Not CMED's.** What a laptop saw while offline. Without a device credential: **401**

Read the shape of that table. The four local ones already work and the deployment does not touch them. The five server ones CMED calls are blocked by **one cause** — no public address — and unblocked by **one act**. Rows 10 to 12 are our recorders talking to our server; they need nothing from CMED.

2 · So what should each CMED team do now?

Team	Can start	Blocked until
Front-end (the page)	**today, on everything** — endpoints 5–8 are live on any PC with the recorder installed	**nothing**
Back-end (the POSTs, the visits GET, the outbox)	**today, writing against document 1**	**testing** needs a hostname from us
Infrastructure	reading this document, and preparing the one config line in section 12	nothing

The browser half is the one with the awkward detail — the gate, the session id, the provisional reply, reading the clinic code off the laptop — and it is completely unblocked. **Nobody on CMED's side needs to wait for our deployment to begin work.**

If CMED's backend team needs to test sooner than our droplet is ready
We can publish a **Cloudflare Tunnel** from the development machine in about fifteen minutes. It gives a real public HTTPS hostname, needs no inbound firewall rule, and costs nothing. **Your code is byte-for-byte identical against the tunnel and against the final droplet** — every endpoint, reply code, key and limit is the same. Only the hostname differs, which is why document 1 asks you to hold it in configuration. **Ask us and we will stand one up.**

PART B

The machine

What we are renting, where, and how fast it answers

3 · The droplet

	Specification	Note
Provider and region	**DigitalOcean, Bangalore (BLR1)**	the lowest-latency region available to Dhaka
Droplet	**Basic / Regular —** `s-8vcpu-16gb`	the smallest standard size meeting every requirement
Processors	**8 vCPU**	the measured load needs about 1.5; the rest is for the busy hour
Memory	**16 GB**	measured use is about 1 GB, capped at 3.4 GB
Disk	**320 GB SSD, included**	**working space only** — see part D
Outbound transfer	**6 TB/month included**	**so uploads to Cloudflare R2 cost nothing**
Operating system	Ubuntu Server 24.04 LTS	supported to 2029
Backups	DigitalOcean droplet backups, enabled	+20%
Price	**$96/month + $19 backups**	plus $45 for storage — part D

4 · Why Bangalore

Region	From Dhaka	Round trip	Verdict
Bangalore — BLR1	~2,400 km	**45–70 ms**	**Chosen**
Singapore — SGP1	~3,000 km	60–95 ms	second choice
Frankfurt — FRA1	~7,400 km	130–170 ms	too far
New York — NYC3	~12,900 km	230–280 ms	too far

Latency does not affect recording, and this is worth understanding

The microphone is driven by the program on the clinic PC, over loopback. **It starts in under a millisecond regardless of where our server is, or whether it is reachable at all.**

The round trip to Bangalore only affects how quickly your two backend messages are acknowledged — and those should be queued on your side anyway, so a slow reply is never visible to a doctor. **If our server is down for an hour, every clinic keeps recording**; each laptop holds about thirteen hours.

5 · What the load actually is

Measure	Value	Note
Clinics	7	fourteen consulting rooms in total
Consultations per clinic day	277	across all rooms
Requests per second, all rooms	**about 1.2**	measured. This is very light
Messages per second from CMED, at peak	**about 0.5**	**no rate limit applies to CMED**
Audio uploaded per clinic day	32.5 GB	277 recordings at ~120 MB

Audio uploaded per month	**844 GB**	**inside the droplet's 6 TB allowance, so free**

PART C

How it is exposed

DNS, the certificate, the ports, and the nine services

6 · The name and the certificate

	Detail
The address	a DNS A record pointing at the droplet — for example `aimscribe.uiu.ac.bd`
Certificate	**Let's Encrypt, issued and renewed automatically** by the gateway. No manual step, no expiry to diarise
TLS	**1.2 and 1.3 only.** Older versions are refused
HTTP	port 80 redirects to HTTPS. It exists only for certificate renewal
HTTP/3	supported, on UDP 443
An IP address will not work	**A certificate cannot be issued for an IP.** The name is a hard prerequisite

Please always use the hostname, never the IP address. Calling the IP directly will fail the certificate check, and the droplet's address can change if it is ever rebuilt.

7 · The security headers on every reply

```
Strict-Transport-Security   max-age=31536000; includeSubDomains
X-Content-Type-Options      nosniff
X-Frame-Options             DENY
Referrer-Policy             no-referrer
Content-Security-Policy     default-src 'none'; frame-ancestors 'none'
Server                      (removed)

Request body cap at the gateway: 2 MB
Application body cap for the clinical endpoints: 1 MB
```

Note the two body caps. The gateway refuses anything over 2 MB outright; the application refuses over 1 MB with a readable 413 `TOO_LARGE`. A normal message is a few kilobytes, so hitting either usually means a file or an image has got into the JSON.

8 · The only two open ports

Direction	Protocol	Port	Source	Why
Inbound	TCP	**443**	all	**the API — this is the one that matters**
Inbound	UDP	**443**	all	HTTP/3
Inbound	TCP	**80**	all	certificate issue and renewal only
Inbound	TCP	22	**AIMS LAB admin IPs only**	administration
Outbound	all	all	—	R2 uploads, OS updates, certificates

No database port, no queue port, no API port is ever exposed. PostgreSQL, Redis and the API itself are reachable only on a private Docker network inside the machine. Administrative access to the databases is by SSH tunnel, never by an open port.

Does CMED need to allow anything outbound?

Only ordinary HTTPS on port 443 to our hostname, from **CMED's server**. If your egress is restricted by destination, that is the one entry to add. **Nothing needs to be opened inbound to CMED**, because we never call you — the traffic is one-directional, CMED to AIMS LAB.

9 · The nine services, and the clinic PCs

Service	What it does	Exposed?
gateway (Caddy)	HTTPS, the certificate, the security headers, the 2 MB cap	**yes — 80 and 443**
api	every endpoint, on internal port 6000	no
postgres	both databases	no
pgbouncer	connection pooling	no
redis	the work queue	no
worker	background processing	no
archive-worker	joins each recording, uploads it to R2, verifies the copy	no
backup	nightly encrypted database dump	no
monitor	watches disk, queue depth and unconfirmed recordings	no

9.1 The clinic PCs need nothing from either infrastructure team

	Detail
What runs there	**AIMScribe.exe**, one per consulting room, installed by AIMS LAB
Listens on	`127.0.0.1:5050` — **loopback only**
Reachable from	**that PC alone.** Not the LAN, not the internet, by design
Firewall changes needed	**none** — on CMED's side or the clinic's
Port forwarding	**none**
Authentication	**no key.** The recorder checks your page's registered web address instead
Outbound from the clinic	HTTPS 443 to our hostname, for uploading recordings

Why no key on the local connection: a secret placed in a web page is readable by anyone who opens the page, so it would protect nothing. The recorder instead accepts connections only from web addresses AIMS LAB has registered, only on loopback, and only from the same machine. **That is why CMED's exact page addresses are a blocking prerequisite** — they are the actual access control.

PART D

Where the recordings live

The droplet disk is working space - the audio is in object storage

10 · Two storages, both needed

The droplet's 320 GB is not the audio archive and is nowhere near enough to be one. Every recording is uploaded to Cloudflare R2 as the raw WAV, encrypted, and removed from the server two days after that copy is verified.

	Droplet disk — 320 GB	Cloudflare R2 — 3 TB
Holds	the OS, the nine services, both databases, logs, and recordings **in transit**	**every recording, permanently, as the original WAV**
How long	two days after its cloud copy is verified (`AIMS_ARCHIVE_LOCAL_RETENTION_DAYS=2`); never before	**permanently.** This is the research dataset
Capacity	two days of recordings, the databases and the rest: a 214 GB budget on the 320 GB disk	**26,200 consultations.** The 18,000-patient study uses 2.06 TB
Cost	included in the droplet	**$45/month**
If it fills	archiving stalls; **clinics keep recording locally**	buy more — $15 per TB per month, no limit

Total monthly cost of the deployment	Monthly	Yearly
Droplet — 8 vCPU / 16 GB / 320 GB, Bangalore	$96	$1,152
Droplet backups	$19.20	$230
Cloudflare R2 — 3 TB	**$45**	**$540**
Transfer to R2 (844 GB/mo, inside the 6 TB allowance)	$0	$0
Domain name	~$1	~$12
Total	**$161**	**$1,934**

R2 charges nothing to read data back, which is why it was chosen over S3 or Spaces: a research dataset is downloaded many times, and on Amazon each full read of 3 TB would cost roughly $250.

11 · The layout, and what Cloudflare can see

```
ON THE DROPLET, in transit
  /srv/aimscribe/archive/catalogue.sqlite3    an index of every session
  /srv/aimscribe/archive/AALO_DHOLPUR/DR0042/2026-10-09/<stem>/
      <stem>.wav             the audio
      <stem>.json            the clinical record CMED sent
      <stem>.manifest.json   proof the audio is intact
      _index.json            what is in this folder

IN CLOUDFLARE R2, permanently
  copies/AALO_DHOLPUR/DR0042/2026-10-09/<stem>.v1.wav.enc
  copies/AALO_DHOLPUR/DR0042/2026-10-09/<stem>.v1.json.enc

  <stem> = P0012345_DR0042_AALO_DHOLPUR_101432_102755_20261009
           patient _ doctor _ clinic _ start _ end _ date
```

The cloud layout mirrors the server layout deliberately — clinic, then doctor, then date — so a recording can be found, or a whole clinic or day restored, without consulting any database.

Every file is encrypted before it leaves our server

AES-GCM, and **the key never leaves AIMS LAB**. Cloudflare can see the folder names and the file names; **it cannot read the audio or the clinical record**. Cloudflare stores bytes it cannot interpret, and so would any future provider — which is what makes putting patient recordings on commodity object storage acceptable at all.

Nothing in R2 is publicly accessible. There is no public bucket, no signed public URL, and no path by which audio reaches CMED or anyone else.

PART E

What changes on CMED's side

One line, and the timing of it

12 · The four places the address lives

	Who	Setting	Changes?
1	Our server	`AIMS_PUBLIC_HOST`	**yes** — ours to do
2	Our archive worker	nothing — it reaches the API inside the machine	**no**
3	**CMED's server**	`AIMS_SERVER_URL`	**YES — this is your one line**
4	Every clinic recorder	`AIMS_BACKEND_URL`	**yes** — ours to do, per PC
—	**CMED's web page**	**nothing — it never knew our address**	**no**

That is the whole of CMED's change: one environment variable on your server. Your page is unaffected because it talks to the recorder on `localhost` and to your own backend — the AIMS LAB address lives only in your backend's configuration, which is exactly where it should be.

13 · The sequence, so nothing is sent into a void

	Step	Who	Note
1	We deploy the droplet and verify it from outside	AIMS LAB	nothing of CMED's is pointed at it yet
2	**We send you the hostname and the key**	AIMS LAB	staging first, production later
3	**You set `AIMS_SERVER_URL` to staging and run the test script**	**CMED**	section 15. Send us the output
4	We move the clinic recorders onto the new address, one first	AIMS LAB	a test consultation on that PC before the other thirteen
5	**Joint end-to-end test**, and we confirm the recording matched and archived	both	**the only check that proves the integration**
6	You switch `AIMS_SERVER_URL` to production	CMED	go live

Messages sent before step 2 are not lost on your side if you queue them — which document 1 asks you to do anyway. But there is no point sending them: nothing will answer. Keep `AIMS_SERVER_URL` unset or pointed at your own stub until we send you the hostname.

13A · The outbox lives on your servers

Each Aalo branch keeps working on its local CMED server through an internet or power cut. **API 2 and API 3 Channel B cannot reach us while the internet is down**, so CMED's software team builds an outbox (document 1, section 16): messages wait in it and go when the connection returns. Late is fine — nothing on our side has a deadline. **Running that outbox reliably is the server team's part.**

	What the server team makes sure of
1	**The outbox is a table on the branch server's own disk**, in the database that survives a power cut, and it is in that server's backups
2	**Whichever server sends to us can reach it outbound**: HTTPS 443 to our hostname, and the key in its secret store. If each branch server sends directly, every branch needs both; if branches sync to a central CMED server that sends, only that one does
3	**It is watched**: the number waiting and the age of the oldest. Alert if the oldest is more than an hour old while the internet is up — that is a stuck outbox, not an outage
4	**A 401 stops it** and pages someone: the key is wrong or revoked
5	**Nothing is purged unsent.** A message is removed only on our 202 or 200

PART F

Health and monitoring

How to tell it is up, and what to do when it is not

14 · The health endpoint

This is the endpoint to monitor. No key, no body, no side effects. It checks the database and the queue on every call.

```
curl -s https://<aims-host>/health

200 {
      "status":   "healthy",
      "database": "connected",
      "redis":    "connected",
      "minio":    "connected",         <-- the segment bucket, checked
      "copy_bucket": "connected",      <-- the 3 TB archive bucket, checked
      "system":   "AIMScribe v3 - AIMS LAB server",
                    ^ assert this STARTS WITH "AIMScribe v3"
      "srs":      "3.x",
      "version":  "3.x.x",
      "mode":     "FastAPI Async"
    }
```

Field	Watch for	Meaning
status	`healthy`	**everything is up.** Anything else is `degraded`
database	`connected`	if disconnected, nothing can be stored. **Alert on this**
redis	`connected`	the queue. If disconnected, archiving stalls
minio, copy_bucket	`connected`	really asked, every 30 s. `refused`, `bucket missing` or `unreachable` mean archiving stops
system	**starts with** `AIMScribe v3`	**see the warning below**
version	changes after a deploy	useful to confirm a release landed

Please assert on the `system` field, not just the status code

There is an older **version 1** AIMScribe backend still running elsewhere, and it answers a `/health` path that looks almost identical. **Pointing at the wrong one costs an afternoon**, because every request succeeds and nothing is ever recorded. The `system` field is first in the response for exactly this reason.

Monitoring guidance	Recommendation
How often	every 30–60 seconds is ample
Alert when	`status != "healthy"` twice in a row, or no answer for 2 minutes
Timeout	5 seconds
Do not	**do not alert the clinics.** A doctor cannot act on it, and recording continues regardless

15 · Proving the gateway, once it is up

	Check	Expected	Proves
1	`nslookup <host>`	the droplet's IP	DNS
2	`curl -sI https://<host>/health`	no TLS warning	the certificate
3	`GET /health`	`status: healthy`	**reachable from CMED's network**
4	the same reply	`system` starts with `AIMScribe v3`	the right system
5	POST API 2 with no key	**401**	auth is enforced
6	POST API 2 with the key	**202**	**the key works**
7	POST the identical body again	**200** ALREADY_RECEIVED	**retries are safe**
8	POST a deliberately invalid body	**422**, naming the bad fields	validation works, and a bad record is **quarantined, not lost**
9	POST a body over 1 MB	**413**	the limit is real
10	GET the visits list for a test patient, with and without the key	**200**, then **401**	the read is authenticated too

Our script does checks 3 to 8 in one command — `python tools/channel_b_test.py --server https://<host> --key <key>`. We will send it with the key. **Run it and send us the output** — it is the evidence both sides need that the gateway is ready.

And then the one check that actually matters. Checks 1 to 8 prove the gateway works; **they do not prove the integration works**. A 202 means stored, not matched. Record one real consultation with all four messages, then **ask us to confirm it was matched and archived**.

Our side, for completeness: an existing database is not migrated by itself. When we deploy this version we apply the two new schema scripts (013 and 014) by hand before the recorders are updated; a new server builds them on first start. Nothing for CMED to do.

16 · When it is down

How long	What happens in the clinics	What CMED should do
minutes	**Nothing.** Recording continues; laptops hold their pieces and retry	**Nothing.** Your outbox retries
an hour	**Still nothing.** Each laptop holds about thirteen hours, and logs what it saw for our dashboard	Nothing. Keep them in the outbox
most of a day	Laptops begin to fill; new recordings eventually stop	Keep them in the outbox. **Do not drop messages**
a message is never sent	**The recording is kept and archived, with no clinical record attached**	**This is the only real failure.** A durable outbox prevents it

Notice what never happens: our server being down does not stop a doctor and does not cut a recording. The system is built so the clinic keeps working whatever we are doing. A message that arrives a day late is completely fine — it attaches to its recording whenever it lands.

What AIMS LAB watches, so you know what we will notice
Every minute: `/health`, from outside the droplet. **Every morning:** the count of unconfirmed recordings — **it should be zero** once outages are over; anything left means a message did not arrive or did not match, and we will ask you to check the outbox. **After any outage:** each laptop's offline period, sent when it reconnects — how long, how many consultations, whether the microphone was fine. **Continuously:** free disk, queue depth, and the upload buffer. **Nightly:** an encrypted database backup, with a restore drill monthly.

PART G

Timeline and summary

What happens when, and the page to keep

17 · The deployment timeline

	What	Who	How long	Blocks CMED?
1	Droplet created, DNS pointed, firewall attached	AIMS LAB	1 hour	no
2	Machine hardened, Docker installed, stack started	AIMS LAB	1 hour	no
3	R2 buckets created, settings filled in, services verified	AIMS LAB	1 hour	no
4	**Clinics and doctors registered, CMED key issued**	AIMS LAB	30 min	**needs CMED's clinic codes**
5	**Hostname and key sent to CMED**	AIMS LAB	—	**unblocks your backend testing**
6	CMED points `AIMS_SERVER_URL` at staging and runs the script	**CMED**	30 min	—
7	Recorders moved onto the new address, one clinic first	AIMS LAB	2 hours	no
8	**Joint end-to-end test and confirmation**	both	1 hour	**yes — both sides needed**
9	Backup restore drill, alerting enabled	AIMS LAB	1 hour	no

Steps 1 to 3 are a single afternoon and need nothing from CMED. The two rows that need you are step 4, which needs your clinic codes, and step 8, which needs a developer on each side for an hour. **Everything else your teams do runs in parallel and is not waiting on us.**

17A · Go-live checklist

	Item	Owner
■	Production hostname and key issued, and stored in CMED's secret store	both
■	**CMED's page addresses registered** (production, staging, preview)	**CMED → AIMS LAB**
■	**Clinic code mapping agreed and loaded**	**CMED → AIMS LAB**
■	`/health` in CMED's monitoring, asserting on `system` **and** `status`	CMED
■	Both POSTs sent from a **durable outbox on the branch server**, retried with backoff, never inline in the request serving the doctor; its depth and oldest age monitored	CMED
■	**The five fields built once per consultation and reused by all four messages**	CMED
■	Clinic code read from the `doctors` command, never hardcoded	CMED
■	Microphone `level` shown on the page	CMED
■	`channel_b_test.py` run, output sent to AIMS LAB	CMED
■	**End-to-end test done, and the recording confirmed matched and archived by AIMS LAB**	both
■	Droplet live in BLR1, backups enabled, both R2 buckets created	AIMS LAB

■	Schema scripts 013 and 014 applied, if the database already existed	AIMS LAB
■	Database backup **restored once** and row counts checked	AIMS LAB
■	Alerting on unconfirmed recordings and on disk above 80%	AIMS LAB
■	All fourteen clinic PCs enrolled against the production hostname	AIMS LAB

The two amber rows are the ones that block us, and both are CMED's to send. The green row is the one that cannot be signed off by either side alone.

18 · One page for the CMED infrastructure engineer

Question	Answer
Where does the server run?	**DigitalOcean, Bangalore (BLR1)**, `s-8vcpu-16gb` — 8 vCPU / 16 GB / 320 GB, Ubuntu 24.04
Why Bangalore?	Lowest round trip to Dhaka, **45–70 ms**
Does latency affect recording?	**No.** The microphone is driven locally over loopback
How do I know it is up?	`GET /health` — no key. Expect `status: healthy`
What else must I assert on?	`system` **starts with** `AIMScribe v3` — an older v1 backend has a similar health path. Do not test for equality
Which ports are open?	**443 TCP and UDP, and 80 TCP** for certificates. Nothing else
Who issues the certificate?	**Let's Encrypt, automatically.** No manual renewal
Can I use the IP instead of the name?	**No.** The certificate is issued for the name
Do I open a port for the clinic PCs?	**No.** The recorder is loopback-only
Does CMED need an inbound rule?	**No.** We never call you
Outbound from CMED?	HTTPS 443 to our hostname, from your server
What changes on CMED's side?	**One line —** `AIMS_SERVER_URL`**.** Your page changes nothing
Where does the audio live?	**Cloudflare R2, 3 TB**, as encrypted WAV. **Not on the droplet's 320 GB**, which is working space
Can Cloudflare read it?	**No.** Encrypted before upload; the key stays at AIMS LAB
Does CMED ever receive audio?	**No. Never**
What if your server goes down?	**Clinics keep recording.** Each laptop holds ~13 hours. Keep the outbox
What if a branch loses internet?	**Recording carries on.** API 2 and 3B wait in the outbox on the branch server and go when it is back. Late is fine
The only real failure?	**A message never sent.** The recording is kept, but with no clinical record
Does a 202 mean success?	**No — stored, not matched.** Ask us to confirm a test recording archived
Total cost?	**$161/month** — $96 droplet, $19.20 backups, $45 storage, ~$1 domain
What do you need from us?	**Your clinic codes** (step 4) and **your page's exact web addresses**

Prepared by AIMS LAB, United International University. **Document 2 of 2**; the companion is "Software Integration", for the team writing the code. Every port, service, variable and limit here is taken from the deployment stack itself. Prices and latency figures are indicative.